



DATA ANALYST PORTFOLIO: YoY Margin Analysis (2014–2017) – Superstore Sales

Alat : Jupiter Notebook, SQL (Explorasi Data & Profiling Data, Data Cleaning, Aggregation), Duckdb SQL, Pandas, matplotlib, seaborn dan sklearn)

Tools : Jupiter Notebook, SQL (Data Exploration & Profiling Data, Data Cleaning, Aggregation), Duckdb SQL, Pandas, matplotlib, seaborn dan sklearn)

Dataset : [Superstore Dataset Final](<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>)
(9,994 rows, 21 Columns)

A. PENDAHULUAN – LATAR BELAKANG BISINIS

Superstore merupakan sebuah jaringan ritel mengalami pertumbuhan penjualan pada periode 2014-2017. Namun, meskipun mengalami bertumbuhan penjuln dan profit, margin yang terlihat mengalami penurunan di tahun 2017.

B. EKSPLORASI & PROFILING DATA

Total Rows	: 9.994
Kolom	: 21
Total Order	: 5.069
Total ID Customer	: 793
Total Customer	: 793
Total ID Product	: 1.862
Total Product Name	: 1.850
Total City	: 531
Total State	: 49
Awal Order	: 2014 -01-03
Akhir Order	: 2017-12-30

Permasalahan Data:

Terdapat selisih antara Product ID dan Product Name yaitu 12. Apabila ingin mengetahui Product ID dan Product Name mana saja yang duplikat atau bermasalah perlu dilakukan cek mendalam pada ke dua kolom ini.

Untuk kolom lainnya tidak di temukan permasalahan data

C. ANALISA YoY & MODELING

Hasil dari Analisa YoY Tahun 2014-2017 terdapat kenaikan Margin di tahun 2016 namun pada tahun selanjutnya adanya penurunan profit dari 13.43 % ke 12.74%, turun 0,69%. Meskipun penurunan margin terlihat kecil dan masih ke dalam status sehat, penurunan ini mesti harus di evaluasi dan di tinjauan faktor apa yang meyebabkan margin ini turun dan kenapa marginnya bisa turun. Penurunan margin ini dapat di lihat jelas berdasarkan grafik visualisasi di bawah ini:

Pertanyaan:

Kenapa margin turun dari 13.43% (2016) ke 12.74% (2017), factor apa yang membuat margin turun, bagaimana cara mengatasi agar margin tidak turun di tahun selanjutnya dan strategi apa meningkatkan margin di tahun selanjutnya?

INTRODUCTION – BUSINESS BACKGROUND

Superstore, a retail chain, experienced saes growth between 2014 – 2017. However, despite this growth in sales and profits, the margin appeared to decline in 2017.

DATA EXPLORATION & PROFILING DATA

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Total Order	: 5.069
Total ID Customer	: 793
Total Customer	: 793
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Total Product Name	: 1.850
Total City	: 531
Total State	: 49
Order Earliest	: 2014 -01-03
Order Latest	: 2017-12-30

Data Issue:

There is a discrepancy of 12 between the Product ID and the Product Name. To identify which Product IDs and Product Names are duplicates or problematic, need to carry out a thorough check of these two columns. No data issue were found in the other column

YOY ANALYSIS & MODELLING

The results of the year-on-year analysis for 2014–2017 show an increase in the margin in 2016; however, in the following year, there was a decline in profit from 13.43% to 12.74%, a decrease of 0.69%. Although the decline in the margin appears small and remains within a healthy range, this decline must be evaluated and the factors causing the margin to fall, as well as the reasons for the decline, must be reviewed. This decline in the margin can be clearly seen in the visualisation graph below:

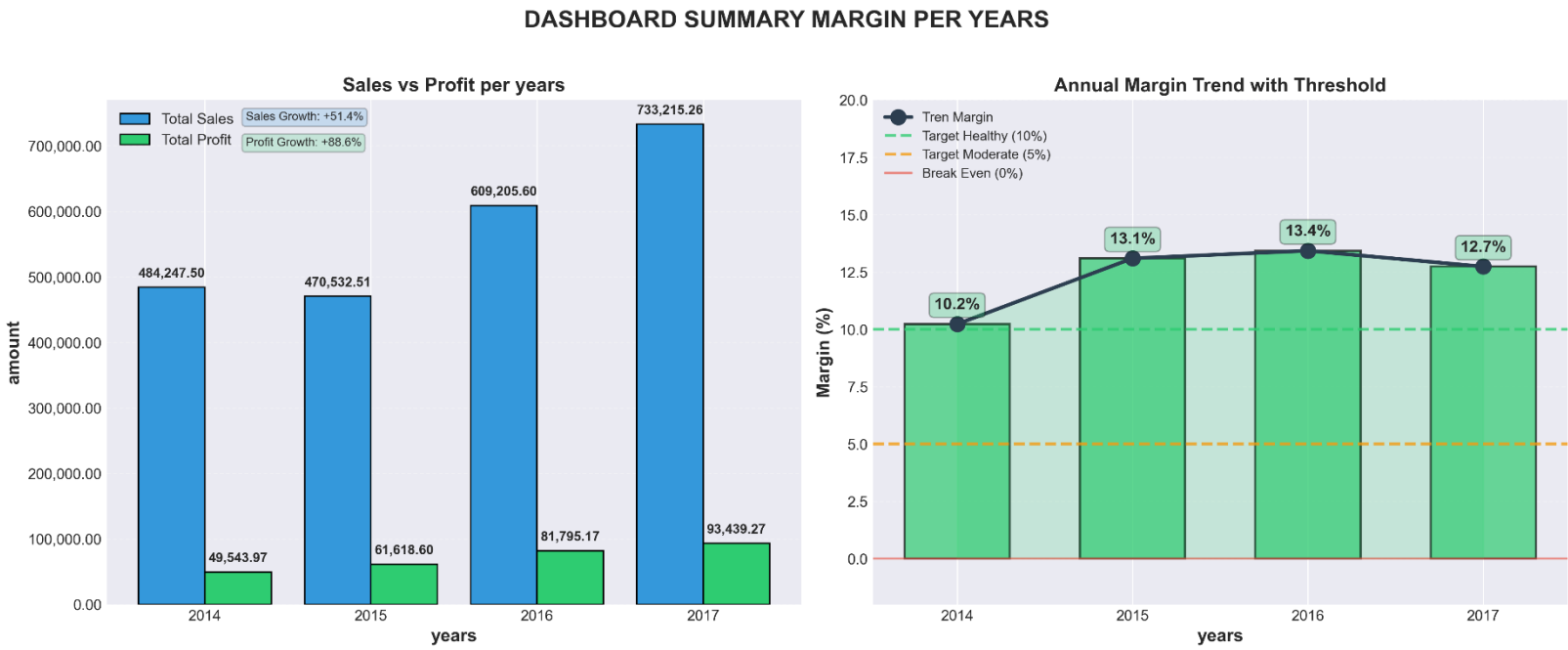
Question:

Why did margin drop from 13.43% (2016) to 12.74% (2017), what factor cause the margin decline, how to prevent margin from declining next year, and what strategies use to increase margins next years?

SUMMARY MARGI REPORT

SUMMARY MARGIN PER YEARS						
Total Sales : 2,297,200.87						
Total Profit : 286,397.01						
Total Average: 12.47%						
TAHUN	TOTAL SALES	TOTAL PROFIT	MARGIN	STATUS	TOTAL MONTH	AVERAGE MONTH
2014	484,247.50	49,543.97	10.23%	Healthy	12	11.50%
2015	470,532.51	61,618.60	13.10%	Healthy	12	12.21%
2016	609,205.60	81,795.17	13.43%	Healthy	12	13.33%
2017	733,215.26	93,439.27	12.74%	Healthy	12	12.84%
The Best Years : 2016 (13.43%)						
The Worst Years: 2014 (10.23%)						

Visualisasi data 1 : Sales vs Profit & Annual Margin Trend



Visualisasi data 2 : Growth Sales, Profit & Margin

